

VeyLance / Interview Sentinel — Consolidated Analysis, Validation & Work Division

Prepared: 2026-10-09 Inputs reviewed:

VeyLance_Interview_Sentinel_System_Level_Research.md, VEYLANCE_24H_MVP_4_PERSON_TEAM_WORK_PLAN.md, plus external source checks (links in Section 10)

0. Short Verdict

The core thesis holds up: **seeing a face on a call no longer proves who is there.**

The weak spots are:

- 1. The files contain no real-world scale numbers (supplied in Section 2).
- 2. Competitor and "open / edge-first" claims need careful wording.
- 3. One attack type, virtual-camera injection, is not covered by the current signals (S1/S2/S4).

1. Problem Statement

Companies verify a candidate's documents, resume and background, but they do **not continuously verify that the person on the live video call is the person being evaluated.**

Real-time face-swap, voice cloning, AI answer-coaching and stand-in "proxy" interviewees now make that gap exploitable. A fake hire can then obtain laptops, credentials and data access.

Core question: How can an organization keep increasing confidence that the person in a remote interaction is real, is the expected person, and is not synthetically represented?

Framing:

Name	Role
VeyLance	Real-time video integrity engine: detection, evidence, risk scoring
Interview Sentinel	Interviewer-facing app: monitoring, alerts, challenge, escalation
Future platform	Continuous identity assurance (interviews, onboarding, help-desk, executive calls)

2. How Big Is the Problem

No single authoritative "number of victims" exists. Present it as a pattern across several sources.

Hiring fraud

- Gartner predicts that by **2028, 1 in 4 candidate profiles worldwide could be fake**. This is a forecast, not a count. In the same survey of 3,000 candidates, **6% admitted to interview fraud**.
- Gartner's 2024 survey found only **7% of recruiting leaders used identity-document validation software**, and 54% agreed AI had made candidate fraud a much bigger problem.
- A Greenhouse survey of 4,136 people found **18% of hiring managers had caught candidates using deepfakes** in video interviews (reported via a secondary source).
- A Palo Alto Networks researcher with no image-manipulation experience built a convincing fake candidate in **about 70 minutes**.

State-backed scheme (North Korean IT workers)

- On **31 July 2026**, the US State Department and FBI, with agencies from 11 countries, issued a joint alert on fraudulent North Korean IT workers.
- US Treasury said these schemes generated **nearly USD 800 million in 2024**.
- The FBI said **8 people were sentenced to prison in 2026 alone** for helping these operations.
- An FBI official said **39 Americans had been arrested or indicted**, and the largest laptop farm found so far held about 100 laptops.
- In July 2026 the FBI disclosed it had identified a North Korean remote IT worker employed by a **US federal agency**.
- The FBI warns that these workers have used AI and face-swapping during video job interviews.

Broader deepfake fraud

- A single deepfake CFO video call cost the engineering firm Arup about **\$25 million**.

Caution: Vendor-sourced statistics (for example GetReal's "41% of firms hired a fraudulent candidate", Pindrop's "1,300% rise") come from companies that sell detection. Present them as "reported by", never as settled fact.

3. Why Today's Checks Fail Inside the Call

1. **Humans cannot spot good fakes.** A meta-analysis of 56 studies put average human detection near 55.5%. An iProov study found only 0.1% of people identified every real and fake clip correctly, and most were overconfident.
 2. **Checks are point-in-time and** perfunctory (done as routine, with little real scrutiny). Document, selfie and background checks happen before or after the call and rely on information the candidate supplied. A reported case passed four video interviews, a photo match and a background check before other controls caught it.
 3. **Liveness checks stop the wrong attack.** They were built for a fake held up to a camera. Neither active nor passive liveness reliably stops injected video, where synthetic frames bypass the physical camera through a virtual camera. iProov reported native virtual-camera attacks up 2,665% across 2024. The World Economic Forum's Cybercrime Atlas (Jan 2026) tested 17 face-swap tools and 8 camera-injection tools against commercial biometric systems. (Injection statistics come mainly from identity-verification vendors.)
 4. **Lab detectors degrade on real call video.** A video-conference deepfake benchmark (VCF) found significant performance loss across 14 detection methods under resolution and compression changes. One aggregator reports losses of up to 50% under real-world degradation.
 5. **Not every attack is a deepfake.** Operators use proxies, laptop farms and teams, and the person a hiring official meets may change depending on the time of day. A face-only detector misses all of that.
 6. **The fallback is costly.** In-person interviews are expensive and exclude some honest candidates.
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4. What VeyLance Solves, and What It Does Not

4.1 What it solves

It watches live video continuously, scores the risk, and tells the interviewer **why**.

Layer	What it checks	Role / limitation
S1 Boundary / blending	Artifacts at the edge of a manipulated face	Fast and explainable; weakened by compression and better generators
S2 Occlusion / landmark stability	Landmark behavior when a hand or object crosses the face	Matches the FBI hand-wave advice; evidence, not proof (real cameras also fail)
S4 Blink / temporal eye behavior	Blink timing, intervals, eye- landmark stability over time	Weak alone; useful as one input to fusion
Fusion	EWMA over S1/S2/S4 to a 0-100 trust score	Prevents one noisy frame from causing an alert

Thresholds (starting values, tune on data): 70-100 HIGH TRUST, 40-69 WARNING, 0-39 ALERT. These are system thresholds, not calibrated probabilities.

Workflow: Detect → Explain → Alert → Challenge → Verify → Escalate

Wording rule: The system says "elevated indicators of synthetic manipulation, additional verification recommended", never "this person is a fraud."

4.2 What it does NOT solve today (say this yourselves)

- Virtual-camera / injection detection (Layer 6, roadmap)
- A real person's face used with a stolen identity (needs identity-consistency, roadmap)
- Voice cloning and A/V sync (S5, roadmap)
- Post-hire access abuse

5. Unique Idea and Competitive Position

Pitch: A transparent, local-first interview-integrity layer that shows signal-level evidence and escalates progressively, instead of an opaque "93% fake" label.

Differentiators

- Explainable per-signal reasons
- Local processing, with no raw video sent to a cloud by default
- Graceful handling of missing signals (no audio means "inactive", not "suspicious")
- Progressive challenge protocol (hand-over-face, head turn, show the room, repeat a phrase)
- Roadmap to two scores: **Authenticity Risk** and **Identity Confidence**
- Privacy-minimal design: discard frames, retain scores and event metadata

Competitors are real (do not claim to beat them on accuracy):

Competitor	What they do	Note
GetReal Security	Real-time identity authentication in live meetings; targets fraudulent job candidates	Closest strategic competitor; also says not every impostor uses a deepfake
Reality Defender	Real-time video / audio detection on calls, marketed for interviews	Mature, multimodal, enterprise
Pindrop	Audio and meeting protection; found deepfake candidates in its own hiring	Strong audio expertise
Sumsub	KYC / liveness with deepfake detection for video identification	Identity-verification platform (not re-checked online)
Intel FakeCatcher	Blood-flow (PPG) signals; claimed 96% accuracy in 2022	Vendor claim, not independent proof

Defensible position: transparency, local deployment, low cost, and an interview-specific workflow. Soften "edge-first" and "open" claims unless you demonstrate them live.

6. Validation of the Research Files

I corroborated (confirmed with additional evidence) the following.

Status	Item
Confirmed	FBI hand-wave guidance and face-swap warning; Palo Alto also recommends hand-over-face, rapid head movement, lighting changes and lip-sync delay checks (supports S2 and a later S5); GetReal, Reality Defender and Pindrop descriptions
Valid but a vendor claim	Intel FakeCatcher 96% (2022 company announcement). Never quote it as independent proof
Partly stretched	NIST Guardians of Forensic Evidence exists, but it is a forensic evaluation program for examiners and evidence, not live calls. Cite it only for "generalization and robustness need testing"
Not re-checked online	Li & Lyu blink paper, Microsoft Video Authenticator, Sumsb launch, FBI Jan 2025 extortion update, EU AI Act page, C2PA, DeepfakeBench
Cannot verify	"27 tests passing" and the working dashboard (repository not visible)
Missing from the files	Real-world scale numbers (added in Section 2) and the injection-attack gap
Self-assessed	The competitor matrix rates your own product; treat it as positioning, not evidence

7. Fix Before Presenting

1. **Pitch honestly:** the MVP detects visual manipulation signals only; identity features are roadmap.
2. **Never claim an accuracy figure** without your own measured test set.
3. **Test under compression:** original clip, then Zoom-like re-encode (720p to 480p), then screen capture. Show the score still moves.
4. **Measure false positives on genuine people:** glasses, low light, head turns, hands near the face.
5. **Rebalance workload:** Person 3 holds S4, fusion, evaluation and escalation. Move the evaluation harness and compression tests to Person 1 after H6.
6. **Prepare an injection answer:** "Virtual-camera detection is Layer 6 on the roadmap; S1/S2/S4 still analyze pixels regardless of source."

8. Work Division (4 People, 24 Hours)

Person	Owns	Must deliver
1: Core / Backend	Camera → OpenCV → MediaPipe → pipeline, signal contract, session state, logging	Stable end-to-end by H6; FPS and latency (average, P95); later the evaluation harness
2: Detection	S1 and S2, scores normalized to 0-1 with confidence and reason text	Unit tests, controlled test cases, limitations slide
3: Temporal / AI	S4, EWMA fusion, trust thresholds, escalation and recovery logic	Configurable thresholds; reproducible metrics (TP, FP, FN, precision, recall, F1, FPR)
4: Frontend / Product	Dashboard, signal cards, alert panel, challenge UI, demo script	Three demo scenarios, recorded backup, screenshots

Shared SignalResult contract

```
json
{
  "signal": "S1",
  "score": 0.0,
  "confidence": 0.0,
  "status": "NORMAL",
  "reason": "Human-readable explanation",
  "timestamp": "2026-10-09T10:00:00"
}
```

- Score: **0** = low suspicion, **1** = high suspicion.
- Trust: **100** = high trust, **0** = low trust.

Timeline

Hour	Required state
H1	Scope and contracts frozen
H3	Modules progressing in parallel
H6	End-to-end MVP
H9	Evaluation running
H12	Hardened MVP + scope freeze
H15	Challenge layer (only if stable)
H18	Demo-ready
H20	Final validation
H22	Presentation-ready, final freeze
H24	Submit

Priorities

- **P0:** camera input, face tracking, S1, S2, S4, fusion, trust score, dashboard, alert state, end-to-end demo
- **P1:** progressive challenge, verification/recovery, metrics, timeline, better explanations
- **P2:** S5 A/V sync, extra signals, more attack scenarios
- **P3 (roadmap only):** CNN models, device fingerprinting, Zoom/Teams/Meet integration, SIEM, C2PA, Kubernetes

Rules

- Reliability beats feature count.
- No major architecture after H12.
- Never fabricate numbers; if the live attack demo fails, use a pre-validated recording and say so.
- Keep the demo deterministic and separate "implemented today" from "roadmap."

9. Golden Demo and Pitch Lines

Demo flow: Genuine session (trust HIGH) → show S1/S2/S4 normal → controlled suspicious event → signals rise → trust drops → explainable alert → challenge → verified (recover) or failed (escalate) → metrics.

Core thesis: "Seeing a person on a screen is no longer proof that the person is present."

One-liner: "We turn the video-call trust boundary into a measurable, explainable security control."

Avoid saying: "99% accurate", "detects every deepfake", "proves identity", "eliminates hiring fraud", "a red alert means the candidate is fake."

10. Sources Consulted

Government / standards

- FBI advisory on North Korean IT worker threats: <https://www.fbi.gov/investigate/cyber/alerts/2025/north-korean-it-worker-threats-to-us-businesses>
- Joint statement on DPRK IT workers, 31 July 2026 (Australian DFAT copy): <https://www.dfat.gov.au/news/media-release/joint-statement-dprk-it-workers>
- Baker McKenzie summary of the 31 July 2026 alert: <https://sanctionsnews.bakermckenzie.com/multiple-governments-issue-joint-alert-on-north-korean-it-workers/>
- NIST Guardians of Forensic Evidence: <https://www.nist.gov/programs-projects/guardians-forensic-evidence>

Scale and hiring fraud

- HR Dive on Gartner's 2028 prediction: <https://www.hrdive.com/news/fake-job-applicants-ai/757126/>
- Greenhouse summary of Gartner's identity-verification report: <https://www.greenhouse.com/de/guidance/mitigate-rising-candidate-fraud-through-identity-verification>
- HR Dive on the Palo Alto 70-minute demonstration: <https://www.hrdive.com/news/fake-job-applicant-deepfake-70-minutes/745924/>
- Palo Alto Unit 42 report: <https://origin-unit42.paloaltonetworks.com/?p=139512>
- Asia Economy on the WSJ investigation (FBI arrest figures): <https://view.asiae.co.kr/en/article/2026081408512551797>
- Asia News Network on the federal-agency case: <https://asianews.network/disguised-north-korean-it-worker-caught-working-for-us-federal-agency-fbi-probes/>

- Pindrop on the 31 July 2026 alert: <https://www.pindrop.com/?p=53305>

Why checks fail

- Injection attacks vs. liveness (vendor sources): <https://www.signzy.com/blogs/liveness-detection-deepfake-injection-attacks> and <https://joinble.io/blog/injection-attacks-liveness-detection-kyc>
- VCF video-conference deepfake benchmark: <https://istina.pskgu.ru/publications/article/806454663/>
- Deepfake detection statistics aggregator (iProov, meta-analysis, Arup): <https://voxbooster.com/blog/deepfake-detection-statistics-2026>

Competitors

- GetReal Security profile: <https://neuronfeed.com/startups/getreal-security>
- Reality Defender on interview deepfakes: <https://www.realitydefender.com/insights/detect-deepfake-job-interviews>
- Intel FakeCatcher (VentureBeat): <https://venturebeat.com/ai/intel-unveils-real-time-deepfake-detector-claims-96-accuracy-rate>

Several figures come from vendors or aggregators. Re-verify any number against its primary source before putting it on a slide.